

AutoInsight AI Research Report

Executive Summary

This dataset contains 891 observations and 12 features. The dataset quality score is 91.9%. The overall reliability score is 95.62, with grade A. Isolation Forest detected 36 anomalous observations.

Dataset Profiling Results

Rows: 891
Columns: 12
Quality Score: 91.9%
Missing Values: 866
Duplicate Rows: 0
Outliers Detected: 36
Outlier Percentage: 5.04%

Dataset Reliability Index

Missing Data Score: 91.9
Duplicate Score: 100
Outlier Score: 94.96
Overall Reliability Score: 95.62
Reliability Grade: A

Model Benchmark Results

Logistic Regression : 69.93%
Decision Tree : 68.53%
Random Forest : 68.53%
Gradient Boosting : 69.23%

Feature Importance

Age : 0.4094
Fare : 0.3917
Pclass : 0.0871
SibSp : 0.0573
Parch : 0.0546

Explainable AI Findings - SHAP

Fare : 0.1223
Age : 0.1186
Pclass : 0.1009
Parch : 0.0404
SibSp : 0.0266

Research Recommendations

- Dataset quality is high and suitable for supervised learning experiments.

- Column 'Cabin' has the highest missingness with 687 missing values.
- No duplicate rows were found, which improves dataset reliability.
- 36 anomalous records were detected and should be reviewed before production use.
- Benchmark multiple models before selecting the final production model.
- Use SHAP explainability to understand how individual features influence model behavior.

Generated by AutoInsight AI - Explainable Dataset Intelligence Platform